

Component Packages

Resistors, Capacitors, Inductors & IC Packages

How to read a datasheet — and pick the right footprint

Audience: Beginner → Intermediate Engineers

Through-hole • SMD • Leadless • BGA • Selection criteria • Assembly

Why Component Packages Matter

Six dimensions in every package decision

Why Component Packages Matter

Footprint

Determines PCB landpattern and placement density

Assembly

Hand-solder vs reflow, pick-and-place feasibility

Thermal

How heat leaves the die — lead frame, pad, or balls

Electrical

Lead inductance, ESR/ESL, parasitic capacitance

Cost

Larger = cheaper to assemble, smaller = cheaper to ship

Reliability

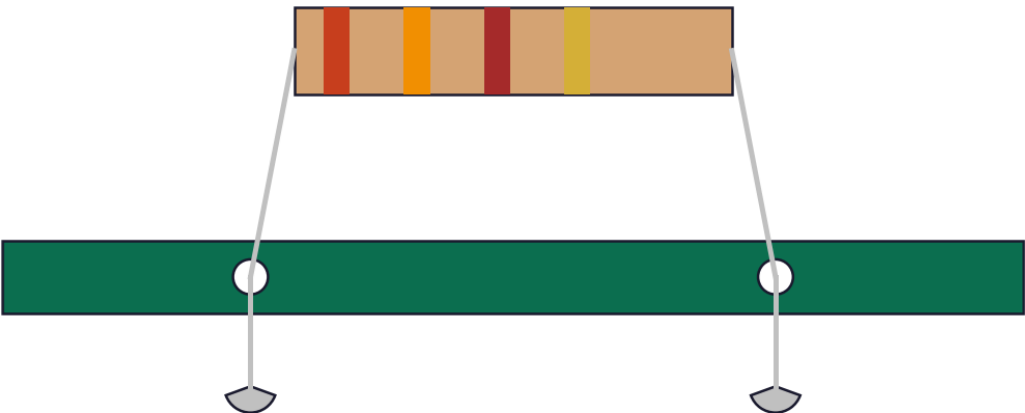
Solder-joint stress, vibration, thermal-cycle endurance

Through-Hole vs Surface-Mount

Two assembly worlds — and where each still wins

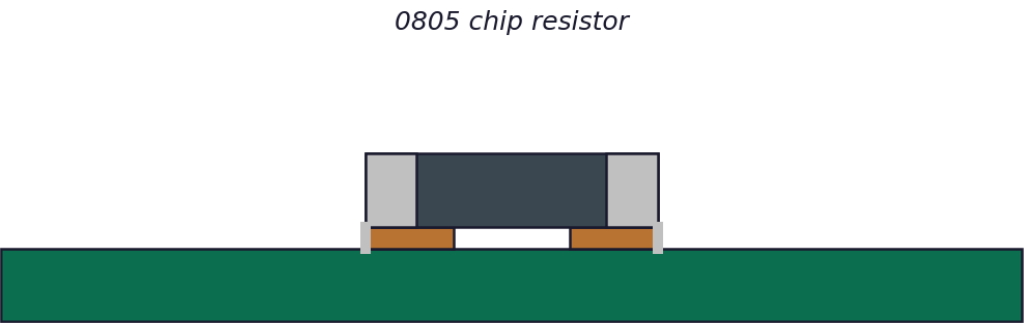
Through-Hole vs Surface-Mount Technology

Through-Hole (THT)



Leads pass through PCB, soldered on the opposite side

Surface-Mount (SMD / SMT)



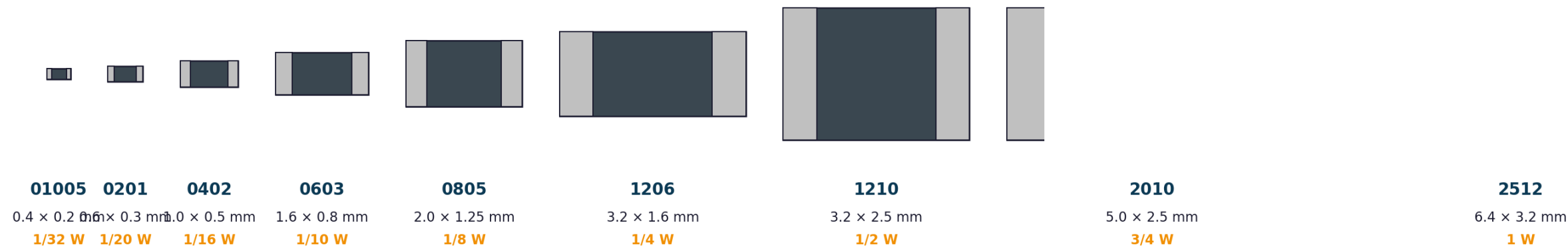
Sits on top of PCB — reflow-soldered onto solder paste

	Through-Hole (THT)	Surface-Mount (SMD)
Mechanical strength	Excellent (lead anchors)	Good (solder only)
Assembly	Wave / hand soldering	Reflow oven, automated SMT
Density	Low	Very high (both sides)
Best for	Connectors, big power, prototyping	Most production: passives, ICs, modules

SMD Chip Sizes — Imperial Codes

0201 / 0402 / 0603 / 0805 / 1206 / 1210 / 2010 / 2512

SMD Chip Sizes (Imperial Codes — drawn to relative scale)



Smaller = denser boards, but harder to hand-solder. 0402 / 0603 dominate consumer & industrial designs; 0201 is mobile-grade.

How to read the code — e.g. '0805' = 0.08" × 0.05" = 2.0 × 1.25 mm

Industrial baseline: 0603/0805. Mobile/wearable: 0402 → 0201. High-power: 1206 → 2512. Metric code exists too (e.g. 0805 = 2012M).

Through-Hole Resistors

Axial bodies, color bands, power dissipation

Through-Hole Resistor Packages



Axial — 1/4 W carbon film

Most common THT resistor



Axial — 1 W metal film

Power dissipation in linear regulators



Wirewound 5 W (ceramic body)

Power supplies, snubbers, dummy loads



TO-220 power resistor

Heatsinkable, 10-50 W

Standard E-series (E12 / E24 / E96) • Tolerances $\pm 5\%$ / $\pm 1\%$ / $\pm 0.1\%$ • TC: ± 50 to ± 100 ppm/ $^{\circ}\text{C}$

Specialty Resistor Packages

Arrays, MELF, current-sense, NTC/PTC, varistors

Specialty Resistor Packages



Resistor Array (R-pack, SIP/MELF) (Metal Electrode Leads) **Current-sense (low Ω , 2512)**

Bus pull-ups, isolated, common-bus Excellent stability, automotive Shunt resistors, m Ω values, 1–3 W



NTC / PTC / Varistor

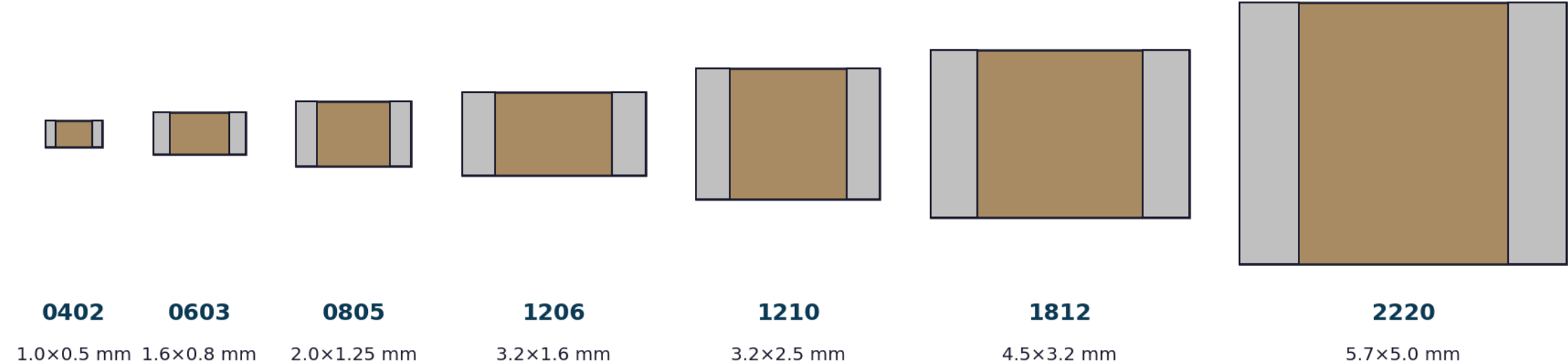
Inrush limiting, surge protection

Tip: pick MELF for high-stability analog, large SMD (2010/2512) for current sensing, and arrays to save BOM lines.

Ceramic (MLCC) Capacitors

The workhorse — 0402 to 2220, COG to Y5V

Ceramic (MLCC) Capacitor Packages



MLCC Dielectric Classes

C0G / NP0: ±30 ppm/°C | ultra-stable, low value (< 100 nF) | filters, timing

X7R: ±15% over -55..+125 °C | general-purpose decoupling, bulk | most common

X5R / X6R: ±15% over -55..+85/105 °C | high-density decoupling

Y5V / Z5U: ±82% over temperature | use only where stability is irrelevant

Tantalum & Polymer Capacitors

Case codes A/B/C/D/E — polarised, low ESR

Tantalum & Polymer Capacitors



A (3216)

3.2×1.6 mm



B (3528)

3.5×2.8 mm



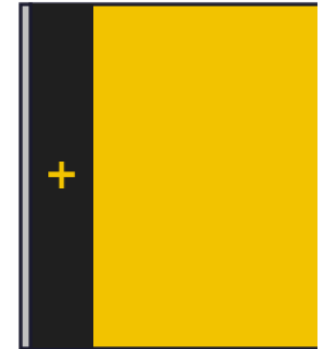
C (6032)

6.0×3.2 mm



D (7343)

7.3×4.3 mm



E (7260)

7.3×6.0 mm

- ▶ Polarised — reverse voltage = failure (sometimes flames!)
- ▶ Choose 2× voltage derating: 3× for polymer tantalum
- ▶ Polymer Ta: lower ESR, safer benign failure mode

▶ Industry alternative to bulky aluminum electrolytics in dense designs

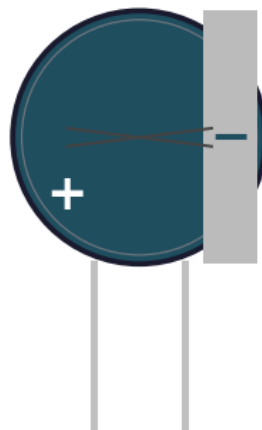
Polarised — orientation matters! • Solid Ta₂O₅ dielectric, MnO₂ or polymer cathode

Polymer tantalum: lower ESR, safer (no ignition failure mode), used in fast-load PSUs

Aluminium Electrolytic Capacitors

Radial / axial / SMD V-chip / snap-in

Aluminium Electrolytic Capacitors



Radial THT

Up to 10 mF, 450 V



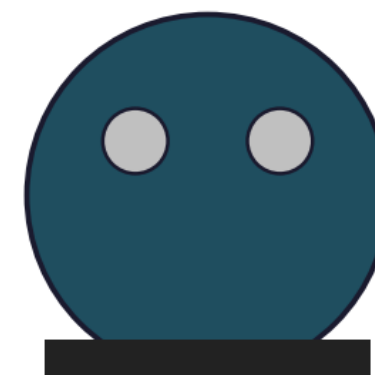
Axial THT

Tube amps, vintage repairs



SMD V-chip

Reflow-compatible alum.



Snap-in / Screw-terminal

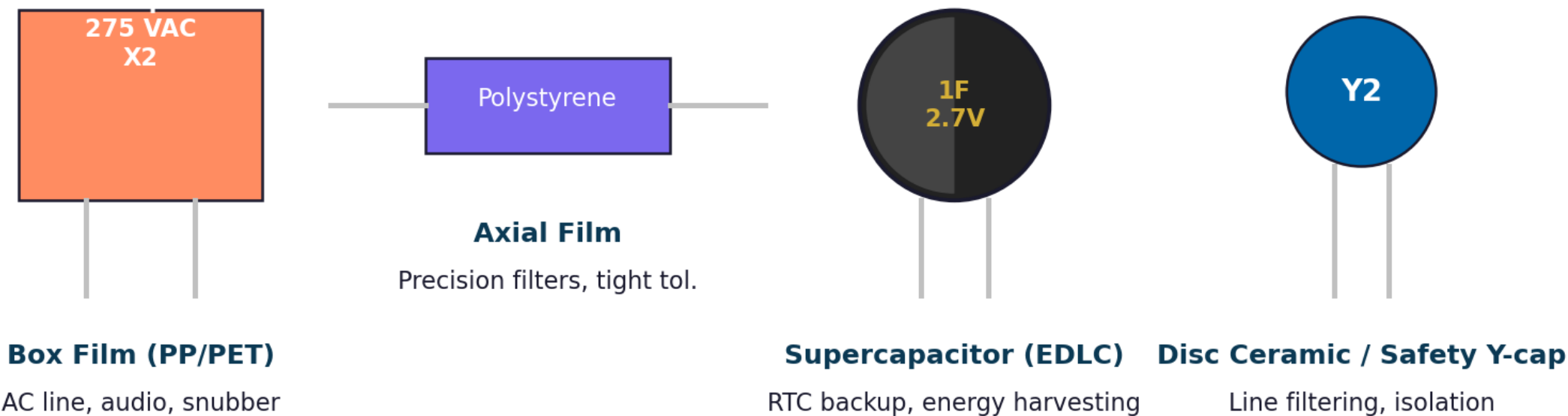
Industrial PSUs, EVs

Polarised • Lifetime halves per +10 °C • Watch ripple-current rating, not just capacitance

Film & Special Capacitors

Polypropylene, polystyrene, supercaps, safety caps

Film & Special Capacitors



Film caps excel in linearity & lifetime; supercaps trade ESR for very large capacitance.

Inductor Packages

Chip, shielded power, toroidal, axial, common-mode chokes

Inductor Packages



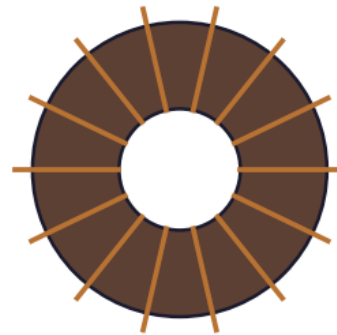
Multilayer Chip (0805/1008)

Ferrite/ceramic, $\leq 100 \mu\text{H}$, RF



Shielded Power Inductor

Buck/boost SMPS, up to 10 A



Toroidal

PFC chokes, line filters



Axial Leaded

Looks like a resistor!



CMC / Transformer

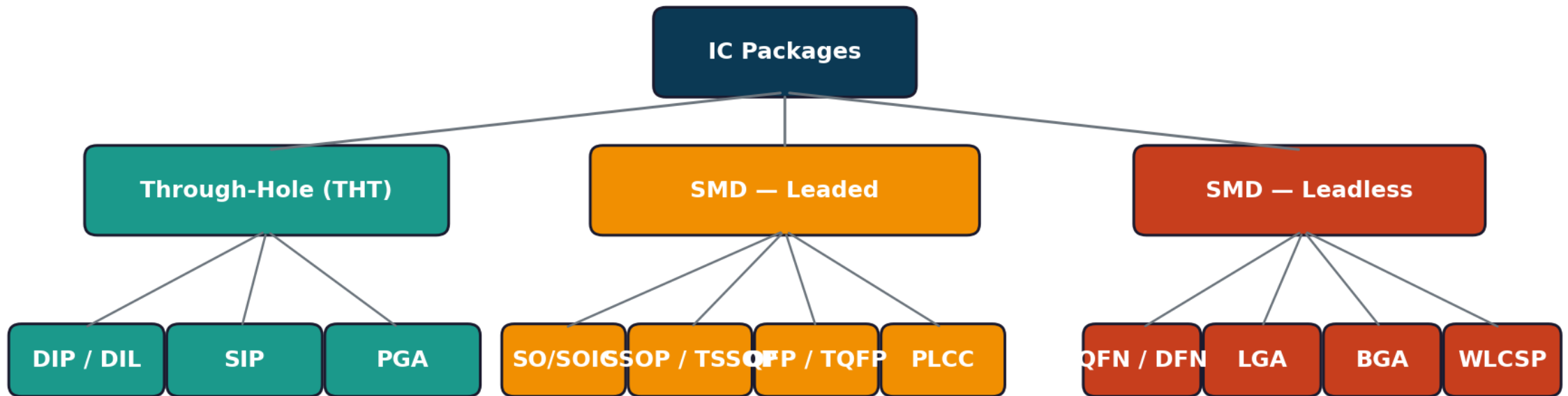
EMI suppression, isolation

Shielded = magnetic field contained (low EMI). Unshielded = cheaper but radiates.

IC Package Family Tree

Through-hole vs SMD-leaded vs SMD-leadless

IC Package Family Tree

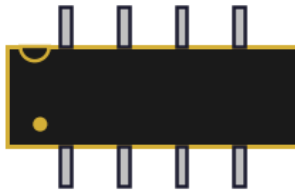


Density & pin count increase → from DIP (≤ 40 pin) to BGA (≥ 1000 pin)

Through-Hole IC: DIP / SIP / PGA / ZIP

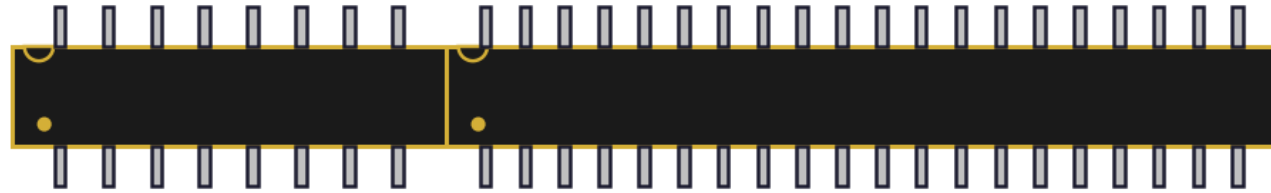
Where you'll still meet plastic dual-inline packages

Through-Hole IC Packages



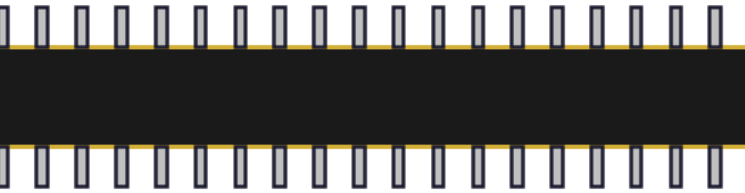
DIP-8 (e.g. 555, op-amps)

2.54 mm pitch, 0.3" wide



DIP-16

74-series logic, EPROM



DIP-40 (ATmega328, 6502 MPU)

0.6" wide variant



SIP (single-inline)

Resistor packs, modules



PGA — Pin Grid Array (legacy CPUs)



ZIP (zigzag inline) — high-density memory

Small-Outline Family: SO / SOIC / SSOP / TSSOP

Gull-wing leaded SMD — easy to inspect, easy to rework

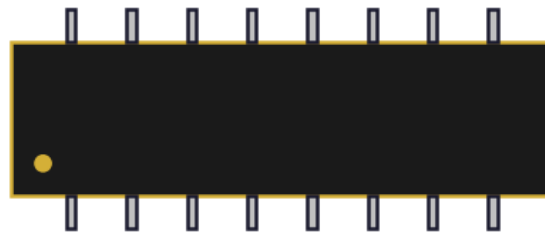
Small-Outline (SO) Family — Gull-wing leaded SMD



SOIC-8 (narrow)

Op-amps, EEPROMs

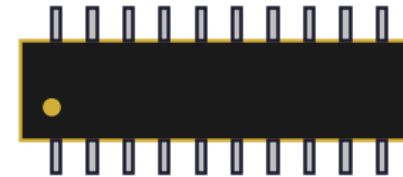
1.27 mm pitch



SOIC-16 (wide)

Logic, interfaces

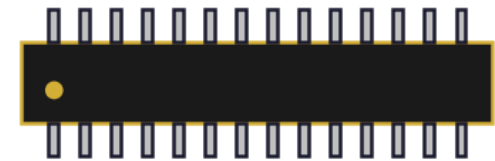
1.27 mm pitch



SSOP-20

Shrunk variant

0.65 mm pitch



TSSOP-28

Thin SSOP, MCUs

0.65 mm pitch

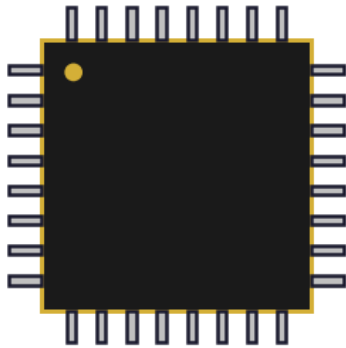
Easiest SMD family to hand-solder

MSOP (mini-SO), SOT-23 (3-lead transistor SMD), TSOP (thin SO for memory) — all share the same gull-wing concept

Quad Flat Pack: QFP / TQFP / LQFP / PLCC

Gull-wing or J-leads on all four sides — pin counts 32 → 256+

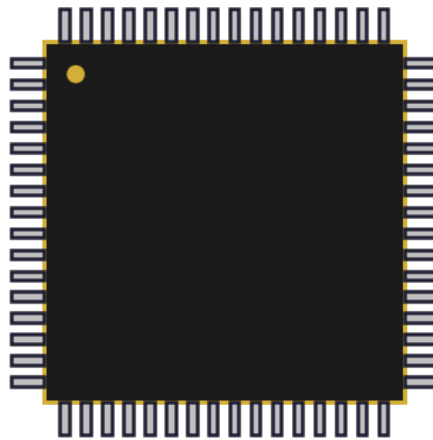
Quad Flat Pack (QFP) Family — leaded on 4 sides



TQFP-32

7×7 mm, 0.8 mm pitch

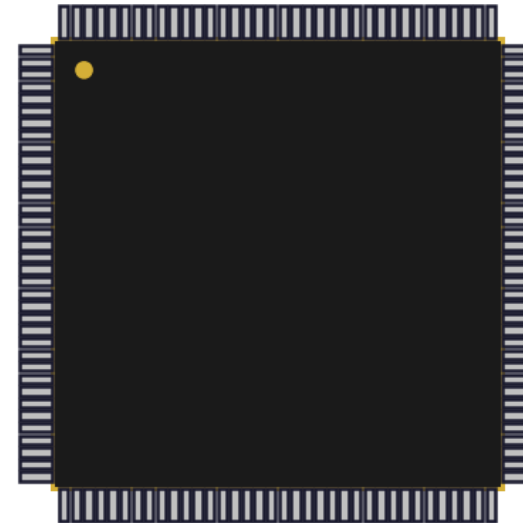
AVR / STM32 entry



LQFP-64

10×10 mm, 0.5 mm pitch

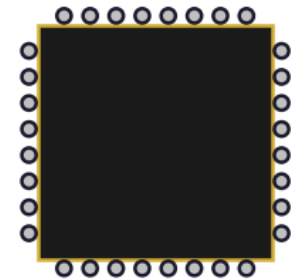
STM32F4 / ESP32 series



LQFP-144

20×20 mm, 0.5 mm pitch

High-end MCUs, FPGAs



PLCC

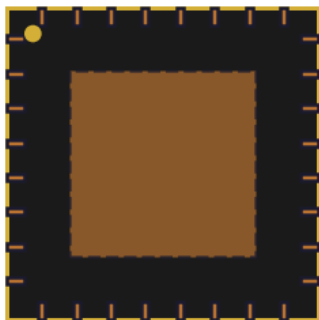
J-leads, socketable
Legacy EEPROMs, BIOS

TQFP = thin, LQFP = low-profile. 0.5 mm pitch is the practical hand-solder limit.

Leadless: QFN, DFN, LGA

Pads underneath — modern default for power-conscious designs

Leadless: QFN, DFN, LGA



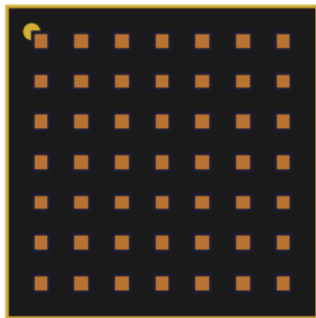
QFN-32

5×5 mm, 0.5 mm pitch
Center pad → thermal/GND



DFN-8 / WSON

2 rows, 0.5 mm pitch
LDOs, MEMS, power switches



LGA (Land Grid Array)

Pads only — no balls
IMUs, RF modules, sockets



Cross-section view

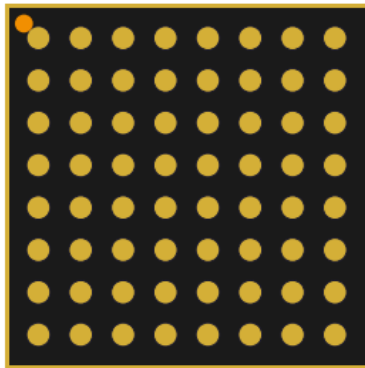
Pads on bottom — no leads to bend

Leadless = lower lead inductance, smaller footprint, but needs proper reflow profile and X-ray inspection.

Ball Grid Array & Wafer-Level CSP

When pin counts cross 200 — and visible joints disappear

Ball Grid Array (BGA) & Wafer-Level CSP



BGA (top view)

Solder balls instead of leads

1.0 / 0.8 / 0.5 / 0.4 mm pitch



BGA cross-section

Solder reflow forms joints

Inner balls = inaccessible (escape routing required)



WLCSP

≤ 0.4 mm pitch, no substrate

Mobile SoCs, ultra-thin devices

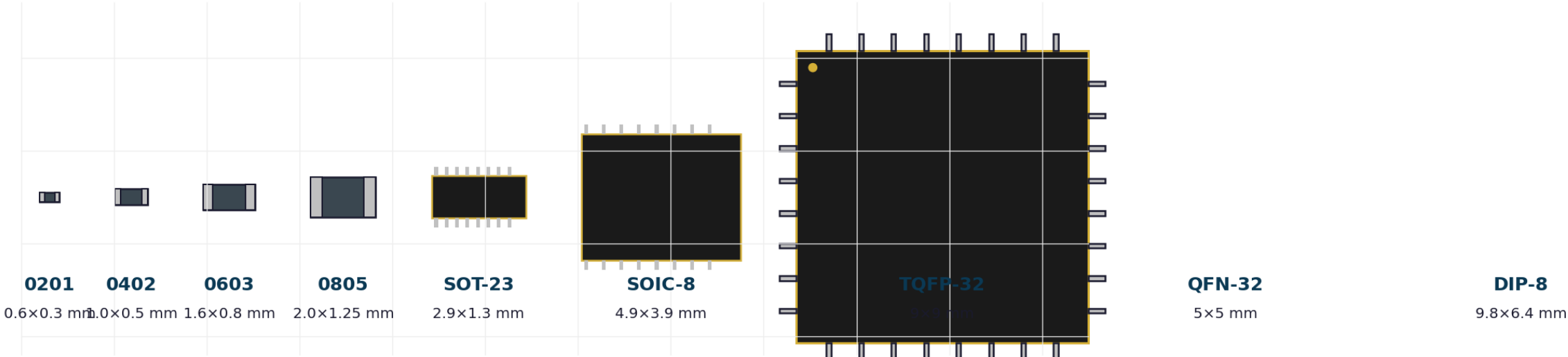
BGA = highest pin density (1000+ pins) — needs ≥ 4 -layer PCB, X-ray inspection, no rework by hand.

Package Size Comparison — Same Scale

From 0201 chip to DIP-40 — drawn at 1 mm = 1 grid unit

Package Size Comparison — Same Scale

(1 grid unit = 1 mm)



From 0.6 mm chip resistor to 10 mm DIP — same boards routinely host the whole range.

How to Choose a Package

Six criteria that drive the decision

How to Choose a Package

Pin Count

More pins → finer pitch / QFN / BGA

Hand-Solder?

0603 / SOIC / TQFP friendly
QFN / BGA need reflow

Board Density

Mobile = WLCSP/BGA;
Industrial = TQFP/QFN

Thermal Load

Look for exposed pad / heatsink tab

Cost & Lead-time

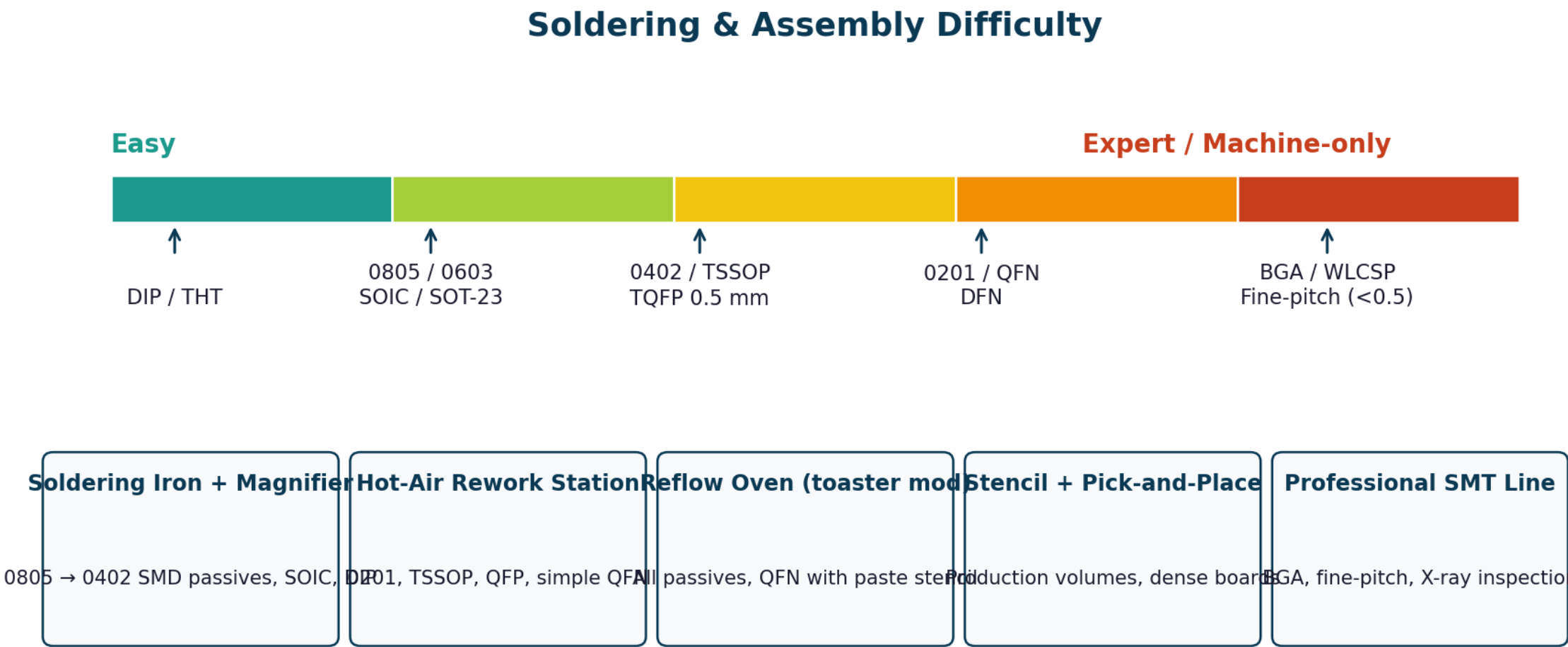
Smaller usually cheaper at volume;
larger easier in prototyping

Reliability / Env.

Automotive / aero prefer leaded
(QFP) over BGA — visual inspection

Soldering & Assembly Difficulty

What you can hand-build vs what needs machines



Quick Reference Cheatsheet

Common packages, pitch, typical pin count, use

Package	Pitch	Pins	Typical use	Hand-solder?
DIP-N	2.54 mm	4–40	MCUs, op-amps, lab/proto	Easy
SOIC	1.27 mm	8–16	Op-amps, EEPROM, drivers	Easy
SSOP / TSSOP	0.65 mm	8–48	Logic, low-pin MCUs	Medium
TQFP / LQFP	0.5 / 0.4 mm	32–256	MCUs (STM32, ESP32), FPGAs	Medium
QFN / DFN	0.5 / 0.4 mm	8–100	LDOs, PMICs, RF, MEMS	Hard
BGA	1.0–0.4 mm	64–2000+	SoCs, FPGAs, RAM, GPUs	Reflow only
WLCSP	≤ 0.4 mm	10–200	Mobile PMICs, sensors	Reflow only
0805 / 0603	—	2	General passives	Easy
0402 / 0201	—	2	Dense / mobile passives	Hard / very hard

Industry Trends in Component Packaging

Where the next 5 years are heading

Where Component Packaging is Heading

1 Miniaturisation

0201 → 01005 → 008004
passives shrinking faster than Moore

2 Integrated Passives

Multi-element arrays,
IPDs replacing 4 discretes

3 Chiplet / 2.5D / 3D

BGA modules with stacked dies
(HBM, Foveros, AMD chiplets)

4 Embedded Components

Resistors/caps inside PCB layers
→ freed surface area

5 Polymer & Hybrid Caps

Replacing aluminium electrolytics
— lower ESR, longer life

6 Lead-Free / RoHS / Halogen-free

Global compliance default

Key Takeaways

Six things to remember from this lecture

Package = Engineering Decision

Footprint, thermal, electrical, assembly, cost — all in one choice.

Default to SMD

Modern boards are SMD-first.
THT only for connectors and heavy power.

Start at 0603 for Passives

Step down to 0402 only if space demands.
Reserve 0201 for production.

Choose Dielectric Wisely

X7R = decoupling. C0G/NP0 = timing/filters.
Never Y5V in a signal path.

Leadless ≠ Lead-free

QFN / BGA hide their solder joints.
Use X-ray or trust your process.

Read the Datasheet — Twice

Verify pitch, body size, thermal pad,
and recommended landpattern.

Questions & Discussion

Ask anything — package choice, soldering, sourcing, datasheets.

References & Further Learning

- ▶ JEDEC.org — official package outline standards (JEP95, etc.)
- ▶ IPC-7351 — generic landpattern standard for SMD components
- ▶ Murata / TDK / Samsung MLCC selector tools — capacitance vs DC-bias data
- ▶ Texas Instruments — package summary tables in every datasheet
- ▶ Eric Bogatin — Signal & Power Integrity Simplified (package parasitics chapter)
- ▶ Phil's Lab (YouTube) — SMD soldering + datasheet reading walkthroughs
- ▶ Application notes from Vishay, Yageo, KEMET on derating and reliability
- ▶ Pick-and-place machine vendor application notes — practical assembly limits

Thank you • Build something this week — even a breakout board.